1 Part 1 - Machines

1.1 Assessment Requirements – Machines

You are required to analyse the mechanical drivetrain of an elevator mechanism (

Figure 1-1) based upon the following elements

- a permanent magnet synchronous motor and a DC link voltage value as an input
- **intermediate twin-ratio gearbox – different from case study**
- counterweight, lift car and load suspended/driven by rope-drum.

The form of the drivetrain is the similar to the one used in the case study in activity sheets but contains an extra gearbox. Note also that you will be analysing a drivetrain with different parameters to the one used in the case study in weeks 01 to 03.

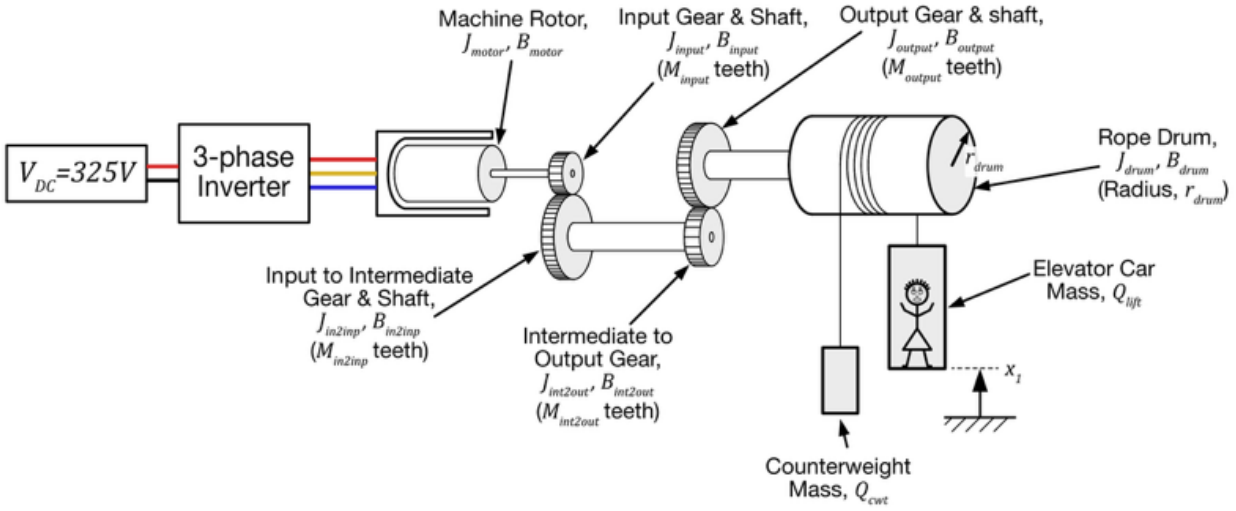


Figure 1-1: Elevator Drivetrain Schematic

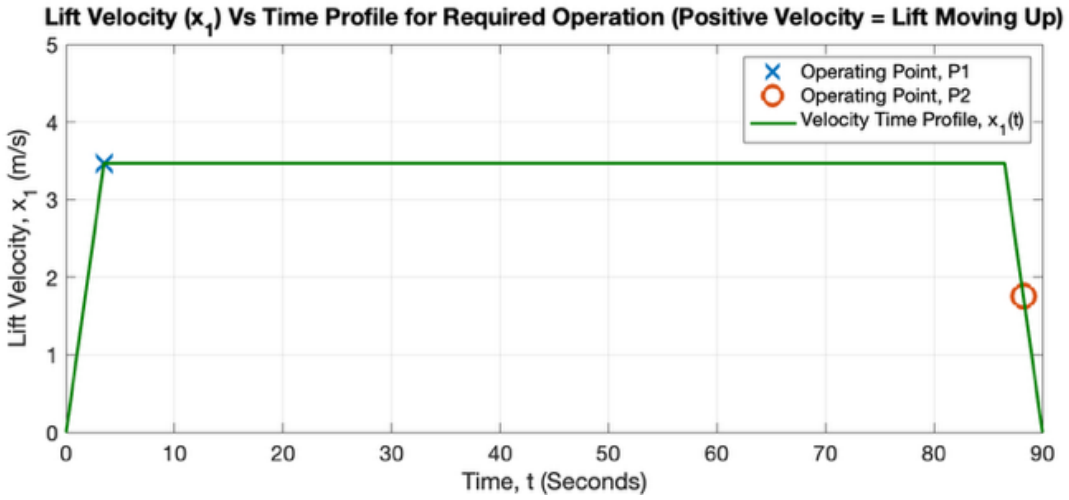


Figure 1-2: Example Speed vs Time Curve (or high level specification)

You will also be provided with a high-level specification (speed vs time curve) that the lift needs to be able to achieve at the output. An example based upon the case study is shown in Figure 1-2.

The parameters of the drivetrain, motor, load and the high-level specification will be provided towards the end of week 08 through blackboard. There are a range of parameter sets, you will be required to undertake the tasks listed below based upon the parameters set you are given. Note, if you use an alternative parameter set to the one you have been assigned you will not pass this aspect of the assessment.

Based upon the high-level specification provided, the task is for you to determine the

(i) Motor angular velocity Vs time profile
(ii) Motor electro-magnetic torque Vs time profile
(iii) Current Vs time profile (d and q axis and total current magnitude)
(iv) Voltage Vs time profile (d and q axis and total voltage magnitude)

and present these using four annotated waveforms similar to those shown in Figure 1-3.

You must also justify the appropriateness of the motor to perform the required loaded speed-time profile based upon the motor specification and the available DC link (3 sentences).

ILOs 1 and 2 marked against:

- correctness of waveforms
- correctness of annotation values for parameter combination
- waveforms (axes labels, white background, >5 cycles etc.),
- clear annotations.

ILO 3 pass awarded for suitable 100-200 word statement that demonstrates clearly how you have derived confidence in your results (i.e. simulations, multiple methods, etc) and elaborates any results you are less confident in and why.

ILO 4 pass awarded for determining correctly if the DC link suggested in the problem could be used successfully with the motor and selected drivetrain and why.

The solution you derived should be uploaded to blackboard along with a simulation that you have used to derive results.

1.2 Example Parameter Set (DO NOT USE THESE)

1.2.1 Lift requirements

The lift must be able to travel a distance of 300m in 90s.

For the purposes of this exercise, you are asked that the speed vs time profile have the form shown below:

- accelerate the lift linearly to the steady-state velocity, $\dot{x}_1^{steady} = 3.5m/s$, from $t = 0s$ to $t = 3.5s$,
- maintain the lift speed at steady-state velocity, $\dot{x}_1^{steady} = 3.5m/s$ from $t = 3.5s$ to $t = 86.5s$,
- decelerate the lift linearly from the steady-state velocity, $\dot{x}_1^{steady} = 3.5m/s$ from $t = 86.5s$ to $t = 90s$.

This results in a linear velocity Vs time profile $x_1(t)$ for the lift that has the form shown in Figure 1-2.

1.2.2 Example Elevator Drivetrain (Load) Parameters

An example of the parameters that describe the details of the load shown schematically in Figure 1-1 are given in Table 1-1. You will be given a set of these for your coursework,

Table 1-1: Mechanical Load Parameters:

Parameter		Symbol	Value	Units
Linear Elevator System	Mass of lift car	Q_{lift}	600	kg
	Mass of "average" person	Q_{person}	80	kg
	Rated capacity of lift	N_{person}	10	people
	Counterweight mass	Q_{cwt}	100% of lift mass + 40% of rated load mass	kg
Rope Drum	Radius	r_{drum}	0.2	m
	Mass moment of inertia	J_{drum}	9.912	kg m^2
	Viscous friction in drum	B_{drum}	10	Nm s rad^{-1}
Output Gear	Mass moment of inertia of gear and shaft	J_{output}	1	kg m^2
	Viscous friction of gear and shaft	B_{output}	0.01	Nm s rad^{-1}
	Number of teeth	M_{output}	60	teeth
Intermediate to Output Gear	Mass moment of inertia of gear	$J_{int2out}$	0.5	kg m^2
	Viscous friction of gear	$B_{int2out}$	0.05	Nm s rad^{-1}
	Number of teeth	$M_{int2out}$	30	teeth
Input to Intermediate Gear	Mass moment of inertia of gear and shaft	J_{in2int}	0.06	kg m^2
	Viscous friction of gear and shaft	B_{in2int}	0.002	Nm s rad^{-1}
	Number of teeth	M_{in2int}	30	teeth
Input Gear	Mass moment of inertia of gear and shaft	J_{input}	0.0123	kg m^2
	Viscous friction of gear and shaft	B_{input}	0.001	Nm s rad^{-1}
	Number of teeth	M_{input}	20	teeth
Input	Supply DC Link Value	V_{DC}	325.3	V_{DC}

1.3 Machine Parameters

An example of the parameters that define the 3-phase, AC, Permanent Magnet Synchronous Machine (PMSM) machine are given in **Table 1-2**. You will be given a set of these for the coursework.

Table 1-2: Machine Parameters

Parameter		Symbol	Value	Units
MACHINE TYPE: 3-phase, AC, Permanent Magnet Synchronous Machine				
Electrical	Stator phase resistance	R_s	0.054	Ω
	Stator d-axis synchronous inductance	L_{ds}	0.00285	H
	Stator q-axis synchronous inductance	L_{qs}	0.00285	H
	Permanent Magnet Flux	λ_{PM}	0.8603	Wb
	Pole Pairs	P	3	
Mechanical	Rated machine rotor rotational speed	ω_R^{rated}	1000	rev/min
	Maximum machine rotor rotational speed	ω_R^{max}	1500	rev/min
	Maximum machine rotor angular acceleration	α_R^{max}	25	rad/s^2
	Mass moment of inertia of machine rotor and shaft	J^{total}	0.25	kg m^2
	Viscous friction of machine rotor and shaft	B^{total}	0.005	Nm s

1.4 Example solution - Machines:

1.4.1 Key attributes (Things we are looking for when determining a mark)

1. Angular Velocity waveform – Annotation labels maximum angular velocity and states angular acceleration at this point
2. Torque speed waveform – Annotation states and highlights the maximum torque requirements AND the mechanical power at this point
3. Current waveforms – Annotation states and highlights the maximum total current magnitude requirements AND the electrical power at this point
4. Voltage waveforms – Annotation states and highlights the maximum total voltage magnitude requirements AND the electrical power at this point
5. Annotate either the voltage OR current graph showing the point at which maximum electrical power is calculation if this is different to either or the points in 3 or 4 above.
6. Correct statement and values chosen from below concerning suitability of DC link
    a. Motor can undertake the task based upon the available DC link and meets all specification requirements.
    b. Motor can undertake the task but NOT based upon the available DC link – Suggest DC link level that would be required
    c. Motor cannot undertake the task based as at some point in the speed time profile the motor maximum ratings are exceeded.
7. Statement (around 100 words) clearly communicating how you have derived confidence in your final results. Try and write this as if you were trying to convince a chief engineer that need to approve or sign off your results.

1.4.2 Demonstration – Machines

1.4.2.1 Analysis/Simulation Outcomes (ILOs 1 & 2)

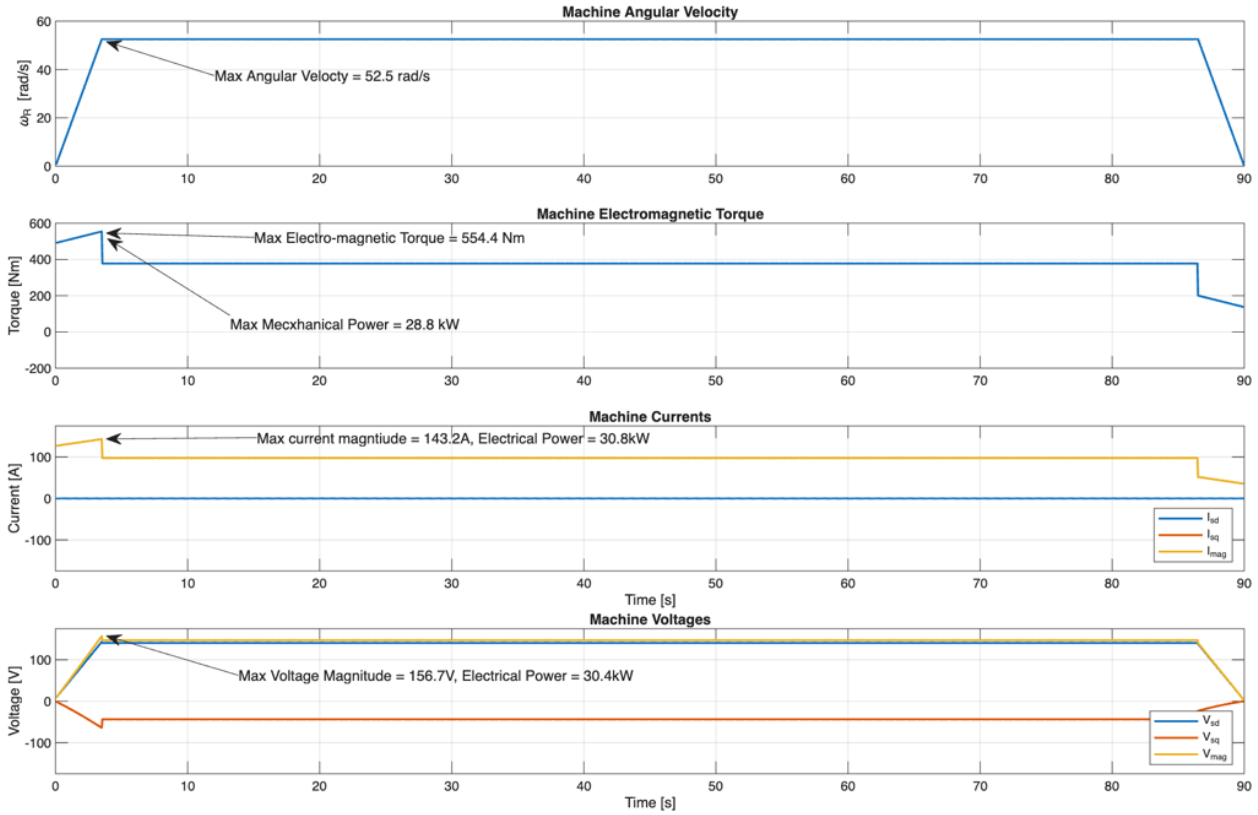


Figure 1-3: Example Figure Submission for Part 1 (not the actual answer)

1.4.2.2 DC Link Suitability (ILO 3 – Pass/Fail)

DC link voltage = XXXV and maximum motor voltage magnitude is XXXV hence the motor and DC link are suitable for the task.

1.4.2.3 Communicate How You Have Derived Confidence in your Results (ILO 4 – Pass/Fail)

I derived confidence in my results using the follow technique(s)/approach(es) … .

I'm less certain in the following results because….